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## A.4 Non-deprecated National Language Alphabet Tables

### A.4.1 Introduction

The national language tables in sections A.3 and A.2 (codes 0x01-0x0D, decimal 1-13) are all deprecated and should never be used.

The subheadings (A.4.10-A.4.24) of this section (A.4), except for section A.4.1, have the third value set to the *hexadecimal* code for the alphabet in question.

Each National Language Alphabet table has three subtables: base table, SS2 table, and SS3 table. The SS3 table is given explicitly only for 0x10 (EUR alphabet). The other SS3 tables are (so far) all the same, very sparse, (only has a few characters), and are given as text comment (in...) rather than a table.

Setting one of the values for National Language Table to a value 0x10 or higher (leaving the other at 0x00) will set all three tables: base table, SS2 table, and SS2 table. If both values are set to a non-zero value, and the first one has a value 0x10 or higher, the second value will be ignored and will also set all three tables consistently.

The default alphabet should not be used for SMS nor CBS, and must not be used for Greek. Use 0x10 and 0x11 instead. The alphabets 0x01-0x0F should never be used.

## A.4.10 SMS/CBS 7-bit European (Latin script) alphabet (0x10) tables

This alphabet for Latin should be preferred over the 7-bit default alphabet (0x00). This (0x10) alphabet supports most European languages that are using the Latin script, and also Vietnamese and Tagalog. For some letters with marks, combining characters are used to represent them. This alphabet should be used instead of the default encoding (0x00) except when the default encoding is explicitly called for, and except for USSD.

### A.4.10.1 SMS/CBS 7-bit European (Latin script)alphabet (0x10) base table

|           |           |           |           | <b>b7</b> | 0          | 0           | 0          | 0         | 1         | 1         | 1         | 1         |
|-----------|-----------|-----------|-----------|-----------|------------|-------------|------------|-----------|-----------|-----------|-----------|-----------|
|           |           |           |           | <b>b6</b> | 0          | 0           | 1          | 1         | 0         | 0         | 1         | 1         |
|           |           |           |           | <b>b5</b> | 0          | 1           | 0          | 1         | 0         | 1         | 0         | 1         |
| <b>b4</b> | <b>b3</b> | <b>b2</b> | <b>b1</b> |           | 0          | 1           | 2          | 3         | 4         | 5         | 6         | 7         |
| 0         | 0         | 0         | 0         | 0         | @<br>0040  | -<br>2212   | SP<br>0020 | 0<br>0030 | i<br>00A1 | P<br>0050 | ç<br>00BF | p<br>0070 |
| 0         | 0         | 0         | 1         | 1         | £<br>00A3  | —<br>005F   | !<br>0021  | 1<br>0031 | A<br>0041 | Q<br>0051 | a<br>0061 | q<br>0071 |
| 0         | 0         | 1         | 0         | 2         | \$<br>0024 | `<br>0300   | "<br>0022  | 2<br>0032 | B<br>0042 | R<br>0052 | b<br>0062 | r<br>0072 |
| 0         | 0         | 1         | 1         | 3         | ¥<br>00A5  | ´<br>0301   | #<br>0023  | 3<br>0033 | C<br>0043 | S<br>0053 | c<br>0063 | s<br>0073 |
| 0         | 1         | 0         | 0         | 4         | è<br>00E8  | ˘<br>0309   | €<br>20AC  | 4<br>0034 | D<br>0044 | T<br>0054 | d<br>0064 | t<br>0074 |
| 0         | 1         | 0         | 1         | 5         | é<br>00E9  | ˙<br>0323   | %<br>0025  | 5<br>0035 | E<br>0045 | U<br>0055 | e<br>0065 | u<br>0075 |
| 0         | 1         | 1         | 0         | 6         | ù<br>00F9  | ˘<br>0303   | &<br>0026  | 6<br>0036 | F<br>0046 | V<br>0056 | f<br>0066 | v<br>0076 |
| 0         | 1         | 1         | 1         | 7         | ì<br>00EC  | ˙<br>0307   | '<br>0027  | 7<br>0037 | G<br>0047 | W<br>0057 | g<br>0067 | w<br>0077 |
| 1         | 0         | 0         | 0         | 8         | ò<br>00F2  | ˘<br>0304   | (<br>0028  | 8<br>0038 | H<br>0048 | X<br>0058 | h<br>0068 | x<br>0078 |
| 1         | 0         | 0         | 1         | 9         | ç<br>00E7  | ˆ<br>0302   | )<br>0029  | 9<br>0039 | I<br>0049 | Y<br>0059 | i<br>0069 | y<br>0079 |
| 1         | 0         | 1         | 0         | A         | LF<br>000A | ˘<br>030C   | *<br>002A  | :<br>003A | J<br>004A | Z<br>005A | j<br>006A | z<br>007A |
| 1         | 0         | 1         | 1         | B         | Ø<br>00D8  | SS2<br>---- | +<br>002B  | ;<br>003B | K<br>004B | Ä<br>00C4 | k<br>006B | ä<br>00E4 |
| 1         | 1         | 0         | 0         | C         | ø<br>00F8  | Æ<br>00C6   | ,<br>002C  | <<br>003C | L<br>004C | Ö<br>00D6 | l<br>006C | ö<br>00F6 |
| 1         | 1         | 0         | 1         | D         | CR<br>000D | æ<br>00E6   | -<br>002D  | =<br>003D | M<br>004D | Ñ<br>00D1 | m<br>006D | ñ<br>00F1 |
| 1         | 1         | 1         | 0         | E         | Å<br>00C5  | ß<br>00DF   | .<br>002E  | ><br>003E | N<br>004E | Ü<br>00DC | n<br>006E | ü<br>00FC |
| 1         | 1         | 1         | 1         | F         | å<br>00E5  | É<br>00C9   | /<br>002F  | ?<br>003F | O<br>004F | Š<br>00A7 | o<br>006F | à<br>00E0 |

LF: LINE FEED; moves to the next line (with implied carriage return).

CR: CARRIAGE RETURN; CR is not used as CR but is used as a filler after the actual SMS/CBS message text. CR should not occur inside a message text, but if it does, it must be converted to LF. Note though that <SS2,CR> is actually CSI (see next table), and that "CR" is not a filler, nor LF.

SS2: SINGLE SHIFT TWO; This code shifts the next 7-bit code unit to refer to the extension of this table (subclause A.4.10.2).

When splitting a message text into submessages, there must be no cut right after an SS2.

U+00DF: Uppercase is U+1E9E (see the extension table).

#### A.4.10.2 SMS/CBS 7-bit European (Latin script) alphabet (0x10) extension (SS2) table

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |           |           |           |                |             |           |           |           |           |           |           |   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|----------------|-------------|-----------|-----------|-----------|-----------|-----------|-----------|---|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |           |           |           | <b>b7</b>      | 0           | 0         | 0         | 0         | 1         | 1         | 1         | 1 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |           |           |           | <b>b6</b>      | 0           | 0         | 1         | 1         | 0         | 0         | 1         | 1 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |           |           |           | <b>b5</b>      | 0           | 1         | 0         | 1         | 0         | 1         | 0         | 1 |
| <b>b4</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>b3</b> | <b>b2</b> | <b>b1</b> | <b>1B</b> | <b>0</b>       | <b>1</b>    | <b>2</b>  | <b>3</b>  | <b>4</b>  | <b>5</b>  | <b>6</b>  | <b>7</b>  |   |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0         | 0         | 0         | 0         | Ö<br>0150      | ª<br>00AA   | σ<br>01A1 | Ą<br>0104 | ı<br>007C | Œ<br>0152 | ¼<br>00BC | Δ<br>26A0 |   |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0         | 0         | 1         | 1         | đ<br>0151      | º<br>00BA   | Ů<br>01AF | ą<br>0105 | Ć<br>0106 | Ě<br>011A | ½<br>00BD | ∞<br>221E |   |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0         | 1         | 0         | 2         | Ů<br>0170      | ǧ<br>011F   | ı<br>01B0 | Ę<br>0118 | ć<br>0107 | Ĝ<br>011E | ¾<br>00BE | ²<br>00B2 |   |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0         | 1         | 1         | 3         | ű<br>0171      | ş<br>015F   | ˘<br>0306 | ę<br>0119 | Ł<br>0141 | Ş<br>015E | “<br>201C | ³<br>00B3 |   |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1         | 0         | 0         | 4         | È<br>00C8      | ˆ<br>005E   | ˙<br>0328 | þ<br>00DE | ł<br>0142 | ě<br>011B | ”<br>201D | ı<br>0131 |   |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1         | 0         | 1         | 5         | İ<br>0069,0307 | Č<br>010C   | ˚<br>0326 | þ<br>00FE | Ň<br>0143 | Ú<br>00DA | ¤<br>00A4 | ú<br>00FA |   |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1         | 1         | 0         | 6         | Û<br>00D9      | č<br>010D   | ˘<br>0327 | Ð<br>00D0 | ń<br>0144 | Í<br>00CD | Ω<br>03A9 | í<br>00ED |   |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1         | 1         | 1         | 7         | İ<br>00CC      | Š<br>0160   | ˘<br>0308 | ð<br>00F0 | Ś<br>015A | Â<br>00C2 | µ<br>03BC | â<br>00E2 |   |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0         | 0         | 0         | 8         | Ò<br>00D2      | š<br>0161   | {<br>007B | Đ<br>0110 | ś<br>015B | Ê<br>00CA | °<br>00B0 | ê<br>00EA |   |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0         | 0         | 1         | 9         | Ç<br>00C7      | Ž<br>017D   | }<br>007D | đ<br>0111 | Ž<br>0179 | Î<br>00CE | ×<br>00D7 | î<br>00EE |   |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0         | 1         | 0         | A         | FF<br>000C     | ž<br>017E   | ˘<br>030B | ≈<br>2248 | ž<br>017A | Û<br>00DB | ÷<br>00F7 | û<br>00FB |   |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0         | 1         | 1         | B         | Ô<br>00D4      | SS3<br>---- | Ý<br>00DD | ≠<br>2260 | Û<br>016E | Ě<br>00CB | ±<br>00B1 | ë<br>00EB |   |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1         | 0         | 0         | C         | ô<br>00F4      | Ǻ<br>0102   | ý<br>00FD | [<br>005B | ű<br>016F | Ĳ<br>00CF | ‰<br>2030 | ï<br>00EF |   |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1         | 0         | 1         | D         | CSI<br>009B    | ǻ<br>0103   | ≤<br>2A7D | ~<br>007E | À<br>00C0 | Ă<br>00C3 | «<br>00AB | ă<br>00E3 |   |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1         | 1         | 0         | E         | Á<br>00C1      | ß<br>1E9E   | ≥<br>2A7E | ]<br>005D | İ<br>0130 | Œ<br>00D5 | »<br>00BB | ö<br>00F5 |   |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1         | 1         | 1         | F         | á<br>00E1      | Œ<br>01A0   | \<br>005C | ·<br>00B7 | œ<br>0153 | Ó<br>00D3 | '<br>2019 | ó<br>00F3 |   |
| FF: FORM FEED; This code is defined as a Page Break character (with implied carriage return). A final recipient of the SMS/CBS message which does not have pagination shall convert FF to LF.<br>CSI: CONTROL SEQUENCE INTRODUCER; enables decimal character references, enables styling. An MS which does not support CSI shall convert it to REPLACEMENT CHARACTER.<br>SS3: SINGLE SHIFT THREE; This code is reserved for the extension to another (SS3) extension table (subclause A.4.10.3), automatically set along with setting this table.<br>When splitting a message text into submessages, there must be no cut right after an SS3. |           |           |           |           |                |             |           |           |           |           |           |           |   |

Though labelled “European”, at least also Turkish, Azeri, Vietnamese, and Tagalog/Filipino can be written using this alphabet.

### A.4.10.3 SMS/CBS 7-bit European (Latin script) alphabet (0x10) extension (SS3) table

|           |           |           |           |           |          |          |          |               |            |                 |            |                |
|-----------|-----------|-----------|-----------|-----------|----------|----------|----------|---------------|------------|-----------------|------------|----------------|
|           |           |           |           | <b>b7</b> | 0        | 0        | 0        | 0             | 1          | 1               | 1          | 1              |
|           |           |           |           | <b>b6</b> | 0        | 0        | 1        | 1             | 0          | 0               | 1          | 1              |
|           |           |           |           | <b>b5</b> | 0        | 1        | 0        | 1             | 0          | 1               | 0          | 1              |
| <b>b4</b> | <b>b3</b> | <b>b2</b> | <b>b1</b> | <b>1B</b> | <b>0</b> | <b>1</b> | <b>2</b> | <b>3</b>      | <b>4</b>   | <b>5</b>        | <b>6</b>   | <b>7</b>       |
| 0         | 0         | 0         | 0         | 0         |          |          |          | NNBSP<br>202F |            |                 | †<br>20BA  |                |
| 0         | 0         | 0         | 1         | 1         |          |          |          |               |            |                 |            |                |
| 0         | 0         | 1         | 0         | 2         |          |          |          |               |            |                 |            |                |
| 0         | 0         | 1         | 1         | 3         |          |          |          |               |            |                 |            |                |
| 0         | 1         | 0         | 0         | 4         |          |          |          |               |            | ƒ<br>0166       |            | ‡<br>0167      |
| 0         | 1         | 0         | 1         | 5         |          |          |          |               |            |                 |            |                |
| 0         | 1         | 1         | 0         | 6         |          |          |          |               |            |                 |            |                |
| 0         | 1         | 1         | 1         | 7         |          |          |          |               |            |                 |            |                |
| 1         | 0         | 0         | 0         | 8         |          |          |          |               | Ĥ<br>0126  |                 | ĥ<br>0127  |                |
| 1         | 0         | 0         | 1         | 9         |          |          |          |               |            |                 |            |                |
| 1         | 0         | 1         | 0         | A         |          |          |          |               |            |                 |            |                |
| 1         | 0         | 1         | 1         | B         |          |          |          |               | IJ<br>0132 | IĴ<br>0132,030B | ij<br>0133 | ĳ<br>0133,030B |
| 1         | 1         | 0         | 0         | C         |          |          |          |               |            |                 |            |                |
| 1         | 1         | 0         | 1         | D         |          |          |          | -<br>2010     |            |                 |            |                |
| 1         | 1         | 1         | 0         | E         |          |          |          |               | Ɔ<br>014A  |                 | η<br>014B  |                |
| 1         | 1         | 1         | 1         | F         |          |          |          | □<br>FFFD     |            |                 |            |                |

In the event that an MS receives a code where a symbol is not represented (including UND) in the above table then the MS shall display the REPLACEMENT CHARACTER (U+FFFD).

U+FFFD: Explicit REPLACEMENT CHARACTER (equivalent in function to SUBSTITUTE, U+001A).

NNBSP: NARROW NO-BREAK SPACE, for recommended space before/after some punctuation in French, as well as space between a number and a fraction (like in 2 ½), a number and %, a number and a unit symbol (like in 40 m), esp. to get a space but still avoid an automatic line break.

U+2010: HYPHEN, a true hyphen, in contrast to U+002D HYPHEN-MINUS. This character is at this position in all of the 7-bit alphabets for SMS/CBS with reference code 0x10 or higher.

U+0133: lowercase 'ij' ligature should be regarded as 'soft-dotted'.

Col. 0-2: The light green marked positions are used for text styling/colouring controls. See section D.

Col. 0-3: These are the same for all 7-bit alphabets with code 0x10 and up, but table not printed here.